# DOXA MOTOR SOLUTIONS

## Car Hire Management System

### Fleet, Booking, Owner Settlement & Automated Communication Platform

## Solution Design Document

**Version 1.0**

| | |
|---|---|
| **Prepared by** | SpectreTech Limited |
| **Prepared for** | Doxa Motor Solutions — Roysambu, Nairobi |
| **Client project** | Doxa Car Hire Management System |
| **Project reference** | SP-DOXA-2026-001 |
| **Purpose** | Solution Design, Scope, Architecture & Delivery Plan |
| **Programme duration** | 6 weeks (27 July 2026 – 7 September 2026) |
| **Date** | 27 July 2026 |

# Document Purpose

This Solution Design Document (SDD) is the single reference that defines what the Doxa Car Hire Management System will do, how it will be built, and in what order it will be delivered. It consolidates the agreed scope, the technology stack, the system architecture, the data model, the user workflows, and the phased delivery plan into one document.

The requirements captured here are drawn directly from the client discovery questionnaire completed by Doxa Motor Solutions, and from the reference system screens reviewed during scoping.

# 1. Overview

Doxa Motor Solutions is a Nairobi-based car hire business operating a fleet of vehicles, some owned directly and some placed on the fleet by third-party car owners. The business currently tracks payments through bank credit alerts and holds records on paper and in Word documents. The platform replaces these manual operations with a connected digital system that handles bookings, dispatch, check-in, lateness penalties, owner settlement, and automated client and owner communication.

**Guiding principle.** The system should not simply store records — it should compute charges, calculate owner payouts, and dispatch the right message to the right person at the right time, so the director can run the business from anywhere with minimal manual work.

## 1.1 Objectives

- Digitise the full hire lifecycle: booking → dispatch → active hire → check-in → settlement.
- Automatically compute hire charges, including lateness penalties, using rules the client configures directly in the system.
- Calculate car owner earnings, deduct agreed expenses, and produce owner statements and payout reports.
- Trigger automated SMS and email messages at each stage — booking, dispatch, payment, reminder, overdue, return, owner payout, and document expiry.
- Prompt clients for M-Pesa payment directly to their phone and reconcile paid against unpaid automatically.
- Track vehicle document expiry (insurance and inspection) with advance warnings.
- Give the director a single dashboard showing fleet status, money collected, outstanding balances, and expiring documents.
- Allow any onboarded client to be found instantly by National ID or phone number.

# 2. Scope

## 2.1 In Scope

### Client & Booking Management

- Client registration capturing full name, phone, email, National ID or passport, copy of ID, driving licence number and copy, physical address, next of kin or emergency contact, employer details, and a second contact person.
- Client lookup by National ID or phone number.
- Suspend and activate client accounts.
- Booking creation, vehicle dispatch, hire extension, and vehicle check-in.
- Dent and condition check recorded at dispatch and at return.

### Fleet & Owner Management

- Vehicle register with type (saloon, SUV / 4x4, van or minibus, pickup, economy) and per-day pricing.
- Car owner records capturing full name, phone, National ID, KRA PIN, logbook copy, bank account details, M-Pesa number, signed agreement, and next of kin.
- Assignment of each vehicle to its owner.
- Owner earnings and expense tracking, with net payout calculation.

- Owner statement and payout report generation.

### Charges, Penalties & Payments

- Automatic hire charge computation per vehicle per day.
- **Client-configurable lateness penalty rules** — the director sets the grace period, penalty rate, and any cap from a settings screen, and the system applies them automatically at check-in.
- M-Pesa Paybill reconciliation and STK Push payment prompts sent directly to the client's phone.
- A running record of who has paid and who is outstanding.

### Document Expiry Tracking

- Insurance and inspection certificate expiry tracked per vehicle.
- Advance warning **7 days** before expiry, as confirmed in the questionnaire.

### Automated Communication

- SMS and email for booking confirmed, vehicle dispatched (with return date and payment details), payment received (with the M-Pesa reference), hire ending soon, vehicle overdue, vehicle returned, owner paid, and document expiry alerts.
- The “hire ending soon” reminder timed at **2 hours before** the end of hire, as confirmed in the questionnaire.
- Email used wherever a client email address exists; SMS otherwise.
- Bulk SMS to clients and broadcast messaging to owners via a registered Sender ID.

### Staff & Access

- Three staff logins across the Director, Office Clerk / Receptionist, Yard Staff, and Accountant roles.
- Role-based permissions enforced server-side.

### Reporting

- Owner payout report, individual owner statement, revenue by vehicle, revenue by month, expenses by vehicle, outstanding client balances, full booking history, client invoices and receipts, and the hire agreement.
- Export in **both PDF and Excel**.

### Director Dashboard

- Cars out on hire now, cars parked and idle, cars due back today, cars overdue, money collected today, money collected this month, who owes money, and documents about to expire.

### Data Migration

- Load the existing vehicle list, car owner list, and client list, currently held in Word documents and on paper.

## 2.2 Out of Scope

The following are excluded unless added by written Change Request:

- GPS vehicle tracking.
- WhatsApp messaging.
- A native mobile application at launch — the web system is mobile-responsive; an app would be a later change.
- Card payment processing.
- Loyalty schemes and promotions.
- Accounting-package integration.
- 24-hour support and infrastructure management after handover.
- Third-party subscription or usage charges — Sender ID, SMS gateway, payment gateway, and email sending costs are borne by the client.

## 2.3 Client Dependencies

Delivery within 6 weeks assumes the client provides, on schedule:

| When | What is required |
|---|---|
| Week 1 | All relevant business details required for development, including company branding assets and existing vehicle, owner and client records for migration. |
| Week 4 | Payment collection details required for the M-Pesa integration. |
| Before go-live | Approved SMS and email message templates, and approved hire agreement wording. |

Further operational details — such as the exact lateness penalty values and the owner commission arrangement — will be gathered at the relevant stage of development, since both are configured directly in the system rather than fixed in code.

*Client-caused delay extends the programme by at least the period of delay.*

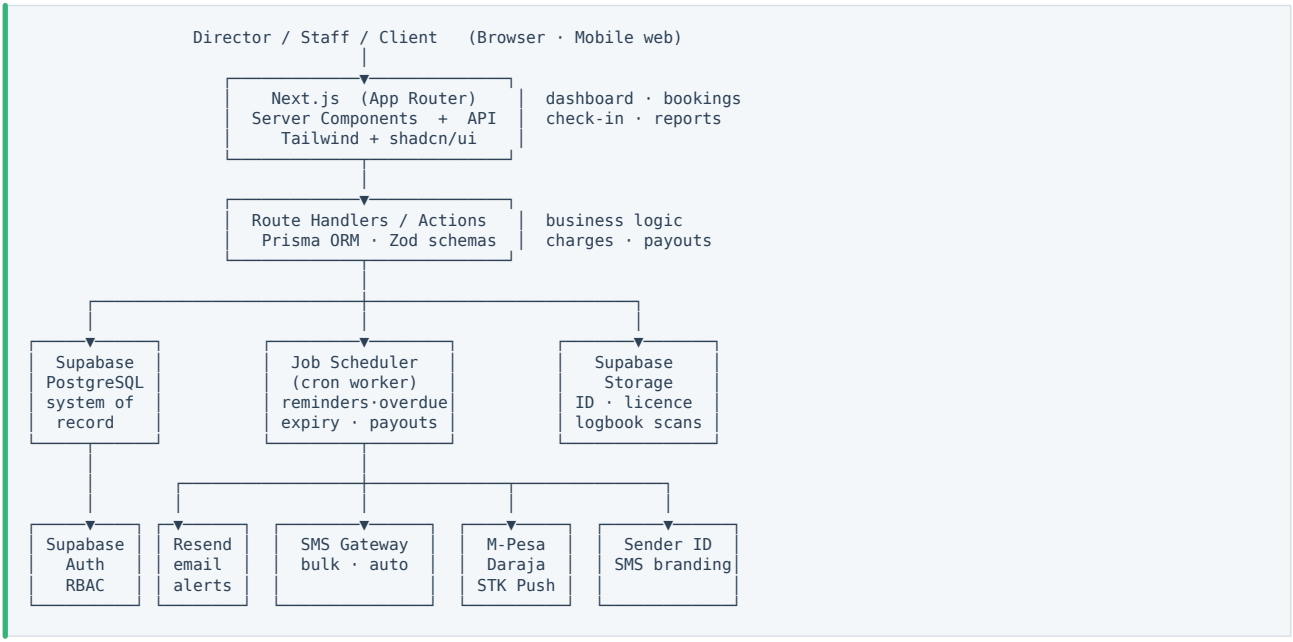
# 3. Architecture

## 3.1 Design Approach

The platform uses **Next.js** (App Router) as a full-stack framework — server components and route handlers provide the API layer, while the client renders the dashboard, booking screens, and reports. **Supabase PostgreSQL** is the relational system of record, accessed through **Prisma** as the typed ORM. Car hire data is highly relational — owners, vehicles, clients, bookings, payments, expenses, and messages all interconnect — so a relational database with JOINs, foreign keys, and transactions gives reliable settlement mathematics and consistent reporting.

**Zod** validates every input at the boundary, in both forms and API handlers, sharing types with the Prisma layer to reduce drift. **Resend** handles branded transactional and bulk email. A dedicated SMS gateway handles Sender-ID SMS. **M-Pesa Daraja** handles STK Push and payment callbacks. A background job runner handles the time-based automation — reminders, overdue checks, and document-expiry alerts — without anyone having to press a button.

## 3.2 Layered Architecture




## 3.3 Layer Responsibilities

| Layer | Responsibility |
|---|---|
| **Frontend**<br>Next.js | Director dashboard, booking, dispatch and check-in screens, owner and client management, and reports. Mobile-first and responsive throughout. |
| **Backend**<br>Route Handlers | Charge and penalty computation, owner payout mathematics, payment reconciliation, message dispatch, and report generation. |
| **ORM**<br>Prisma | Typed data access to PostgreSQL, versioned migrations, and transactional settlement writes. |
| **Validation**<br>Zod | Schema validation of all form and API input, with shared inferred types. |
| **Database**<br>Supabase PostgreSQL | System of record for all relational data — JOINs, transactions, and foreign keys for reliable reporting. |
| **Auth**<br>Supabase Auth | Staff authentication and role-based access control. |
| **Storage**<br>Supabase Storage | Scanned IDs, driving licences, logbooks, signed agreements, and the company logo. |

| Layer | Responsibility |
| --- | --- |
| **Scheduler**<br>Cron worker | Time-based automation: hire-ending reminders, overdue detection, document-expiry alerts, and scheduled owner statements. |
| **Email**<br>Resend | Branded transactional and bulk email with tracked delivery. |
| **SMS**<br>Gateway + Sender ID | Automated and bulk SMS branded with the company Sender ID. |
| **Payments**<br>M-Pesa Daraja | STK Push prompts and payment callbacks for reconciliation. |

# 4. Technology Stack

| Layer | Technology | Role |
|---|---|---|
| Framework | Next.js (React + TypeScript) | Full-stack: SSR UI, dashboards, and API route handlers |
| UI | Tailwind CSS + shadcn/ui | Consistent branded component library in blue and white |
| ORM | Prisma | Typed database access and versioned migrations |
| Validation | Zod | Input validation and shared types |
| Database | Supabase PostgreSQL | Relational system of record |
| Authentication | Supabase Auth | Staff login and role-based access control |
| Storage | Supabase Storage | ID, licence and logbook scans, agreements, logo |
| Email | Resend | Branded transactional and bulk email |
| SMS | SMS gateway + registered Sender ID | Automated and bulk branded SMS |
| Payments | M-Pesa Daraja | STK Push and Paybill reconciliation |
| Automation | Scheduled cron functions | Reminders, overdue checks, expiry alerts, statements |
| Reports | Server-side PDF + Excel export | PDF and Excel report generation |
| Deployment | Vercel + Supabase | Managed hosting for application and data |
| Monitoring | Sentry | Error tracking |

# 5. Roles, Permissions & Access Control

## 5.1 Permission Model

Access is role-based and enforced server-side, never in the user interface alone. The questionnaire confirmed **three staff logins** across the roles set out below. Roles and their exact permissions are configurable, so the structure can adapt as Doxa grows.

## 5.2 Starting Roles

| Role | Typical Responsibilities |
|---|---|
| **Director / Owner** | Sees everything: full dashboard, all money, all reports, owner settlement, staff management, message controls, and system settings including penalty and commission configuration. |
| **Office Clerk / Receptionist** | Registers clients, creates bookings, dispatches vehicles, and generates receipts. |
| **Yard Staff** | Releases and receives vehicles, and records dent and condition checks at dispatch and return. |
| **Accountant / Bookkeeper** | Records expenses, reconciles payments, and generates owner statements and financial reports. |

# 6. Core Workflows

## 6.1 Booking and Dispatch Workflow

Based on the process described by the client:

- A client requests a car and staff **verify the driving licence validity**.
- The client is asked to visit the office for **vetting**.
- On successful vetting, staff complete **documentation** — the client record is captured, or looked up by National ID or phone.
- The client **makes payment** via an M-Pesa STK Push prompt or Paybill; the system reconciles the payment and marks the booking paid.
- Staff record the **dent and condition check** on the vehicle.
- The vehicle is **dispatched**. The system sends the client the dispatch details — registration, hire period, return date and payment details — and notifies the Director with the dispatch, the payment received, and the remaining fleet count.

## 6.2 Return and Check-in Workflow

- The vehicle returns and yard staff **verify there are no new dents** beyond those recorded at dispatch.
- Staff **check that all components** are working.
- Staff mark the vehicle **checked in**; the system computes final charges including any lateness penalty.
- The system sends the client a **return confirmation** and updates fleet status.

## 6.3 Charge & Penalty Computation

- Base charge equals the per-day rate multiplied by the number of days.
- A lateness penalty is applied automatically when the actual return exceeds the agreed end time.
- **Penalty rules are configured by the client in the system, not fixed in code.** From a settings screen the Director defines the grace period, the penalty rate (per hour or per day), and any maximum cap. Rules can be changed at any time and apply to bookings from that point forward.
- Every applied penalty records which rule and which values produced it, so the calculation is always traceable and can be explained to a client.

## 6.4 Owner Settlement Workflow

- For each owner and period, the system totals **earnings** from that owner's vehicles.
- Agreed **expenses and deductions** are subtracted — for example service, parts, tracking, or a recurring weekly deduction.
- **Net pay** equals earnings less expenses, as shown on the owner payout report.
- The system generates the owner statement or invoice and sends an owner notification when payment is made.

As with penalties, the commission and deduction arrangement is configurable per owner in the system — a percentage split, a fixed recurring deduction, itemised expenses, or a combination — so different owner agreements are supported without code changes.

## 6.5 Automated Communication Workflow

A scheduled worker and event triggers dispatch messages:

| Trigger | Channel | Message |
|---|---|---|
| Booking created | SMS + email | Booking confirmed |
| Vehicle dispatched | SMS + email | Dispatch details, return date and payment information |
| Payment received | SMS + email | Payment received, with the M-Pesa reference |
| 2 hours before end of hire | SMS + email | Hire ending soon |

| Trigger | Channel | Message |
|---|---|---|
| Past the end time | SMS + email | Vehicle overdue |
| Vehicle returned | SMS + email | Return confirmation |
| Owner paid | SMS + email | Owner payout notification |
| 7 days before document expiry | SMS + email | Insurance or inspection expiring |

Email is used wherever a client email address exists; SMS is always available via the Sender ID. Bulk SMS to clients and broadcasts to owners are supported.

### 6.6 Document Expiry Workflow

The scheduler checks vehicle insurance and inspection dates daily, raises a **7-day** advance alert to the Director, and surfaces the item on the dashboard.

# 7. Data Model

Core relational entities in Supabase PostgreSQL, defined as Prisma models with Zod validation:

| Entity | Key Fields |
|---|---|
| owners | id, full_name, phone, national_id, kra_pin, logbook_url, bank_details, mpesa_number, agreement_url, next_of_kin, commission_config, created_at |
| vehicles | id, owner_id, reg_no, description, type, daily_rate, status, insurance_expiry, inspection_expiry |
| clients | id, full_name, phone, email, id_or_passport_no, id_copy_url, licence_no, licence_copy_url, address, next_of_kin, employer, second_contact, status, created_at |
| bookings | id, client_id, vehicle_id, start_date, end_date, days, base_charge, status, created_by, created_at |
| condition_checks | id, booking_id, stage (dispatch / return), dents_notes, components_ok, checked_by, created_at |
| penalty_rules | id, name, grace_period, rate, rate_unit (hour / day), cap, active, updated_by, updated_at |
| penalties | id, booking_id, rule_id, hours_late, amount, calculation_notes |
| payments | id, booking_id, amount, method, mpesa_ref, status, paid_at |
| expenses | id, vehicle_id, owner_id, description, amount, period, created_at |
| owner_payouts | id, owner_id, period, gross_earnings, total_expenses, net_pay, status, paid_at |
| messages | id, recipient_type (client / owner), recipient_id, channel (sms / email), template, status, sent_at |
| users | id, name, email, role, active, created_at |
| documents | id, vehicle_id, type (insurance / inspection), expiry_date, alert_sent |

The relational structure enables JOIN-based reporting — revenue by vehicle, revenue by month, owner statements, outstanding balances, and full booking history.

# 8. Security & Data Protection

- Staff authenticate through Supabase Auth, with role checks enforced server-side.
- All traffic is served over HTTPS.
- Scanned IDs, licences, logbooks and agreements are stored in access-controlled Supabase Storage, never in public buckets.
- Payment gateway credentials and API keys are held as server-side secrets, never in the repository or the client bundle.
- Doxa Motor Solutions is the data Controller of client and owner personal data; SpectreTech Limited acts as Processor on documented instructions.
- Client lookup by National ID or phone number is restricted to authenticated staff roles.
- Changes to penalty and commission settings are recorded with the user and timestamp for accountability.

# 9. Delivery Plan

## 9.1 Plan at a Glance — 6 Weeks Total (27 July – 7 September 2026)

| Phase | Focus | Duration | Dates |
|---|---|---|---|
| **1. Setup & Foundation** | Repository, environment, database schema, authentication, branded base UI | Week 1 | 27 Jul – 2 Aug |
| **2. Core Build** Clients, Fleet, Owners | Client, owner and vehicle registers; lookup; document storage | Week 2 | 3 Aug – 9 Aug |
| **3. Bookings, Dispatch & Charges** | Booking, dispatch, check-in, condition checks, charge engine and configurable penalty rules | Week 3 | 10 Aug – 16 Aug |
| **4. Payments & Owner Settlement** | M-Pesa STK Push, payment reconciliation, owner payout calculation, statements | Week 4 | 17 Aug – 23 Aug |
| **5. Automation & Reporting** | SMS and email automation, scheduler, director dashboard, PDF and Excel reports, data migration | Week 5 | 24 Aug – 30 Aug |
| **6. Testing, Training & Deployment** | End-to-end testing, hardening, staff training, go-live | Week 6 | 31 Aug – 7 Sep |

**Key point.** The six-week window is tight and assumes client-side information, branding assets, existing records, and payment details arrive on time. Any delay in those dependencies moves go-live accordingly.

# 10. Project Cost & Payment Schedule

## 10.1 Cost Basis

| Item | Detail |
|---|---|
| Currency | Kenya Shillings (KSh) |
| Total project cost | **KSh 40,000** |
| Payment method | M-Pesa |
| Invoice / evidence | Yes — M-Pesa receipt issued per payment |

## 10.2 Payment Schedule

| # | Milestone | When | Amount |
|---|---|---|---|
| 1 | Project commencement | Week 1 — 27 July 2026 | KSh 10,000 |
| 2 | Core build underway | Week 2 | KSh 10,000 |
| 3 | Development and deployment complete | Week 6 — 7 September 2026 | KSh 20,000 |
| | **Total project cost** | | **KSh 40,000** |

## 10.3 Acceptance & Change Control

- **Review period:** each submitted milestone may be reviewed within 5 Business Days.
- **Rejection:** a rejection should be written and itemised, identifying a material failure against an agreed acceptance criterion.
- **Rectification:** valid material failures are addressed within 10 Business Days and resubmitted.
- **Deemed acceptance:** if no compliant written rejection is received within the review period, the milestone is deemed accepted. Minor defects that do not prevent operational use do not delay acceptance and are addressed during testing or the post-launch support period.

- **Acceptance criteria:** all agreed functionality delivered; no critical defects; production deployment successful; documentation delivered.
- **Revisions:** up to three consolidated revision rounds are included within scope. A new module, integration, role structure, workflow, or provider is a Change Request, not a revision.
- **Change approval route:** written approval from an authorised SpectreTech Limited representative.

## 11. Deployment, Handover & Support

- Deployment to the agreed production environment: release preparation, environment configuration, launch checks, domain and DNS assistance, and handover guidance.
- On full payment, SpectreTech Limited hands over project-specific source code, database access or an agreed export, admin and deployment credentials, and technical documentation.
- Ongoing server administration, infrastructure monitoring, backups, domain renewal, Sender ID, SMS, payment gateway and email running costs, and 24-hour cover are excluded unless separately contracted.
- Post-launch support runs for 60 calendar days from go-live, covering reproducible defects in delivered functionality and deployment issues attributable to the delivered code. New features and redesign are excluded.

## 12. Deliverables Summary

| | | | |
|---|---|---|---|
| **1** | Director Dashboard | **11** | M-Pesa Payment Integration |
| **2** | Client Management & Lookup | **12** | Payment Reconciliation |
| **3** | Owner Management | **13** | Owner Payout & Statements |
| **4** | Vehicle / Fleet Register | **14** | SMS Automation (Sender ID) |
| **5** | Booking & Dispatch | **15** | Email Automation |
| **6** | Check-in & Condition Checks | **16** | Bulk SMS & Owner Broadcast |
| **7** | Charge Engine & Penalty Rules | **17** | Document Expiry Alerts |
| **8** | Role-Based Access Control | **18** | PDF & Excel Reports |
| **9** | Document / File Storage | **19** | Data Migration |
| **10** | Authentication System | **20** | Deployment, Training & Source Code |

## 13. Brand & Visual Direction

The user interface carries Doxa Motor Solutions' brand identity: **blue and white**, as seen in the company logo. The digital experience should read as a natural extension of the Doxa brand rather than a generic third-party tool.

## 14. Assumptions & Risks

| Item | Note |
|---|---|
| **Timely client information** | The six-week schedule assumes branding assets, existing records, payment details, and approvals arrive on time; delays extend the programme. |
| **Configurable business rules** | Penalty and commission rules are set by the client in the system. Accuracy of charges and payouts depends on the client configuring these correctly and keeping them current. |
| **Third-party services** | SMS deliverability, email deliverability, and M-Pesa availability depend on provider capabilities and are not guaranteed to be complete. |
| **Sender ID lead time** | Registration of the SMS Sender ID depends on the mobile operators and may affect the branding of automated messages at go-live. |
| **Data migration quality** | Existing records are held in Word documents and on paper; migration accuracy depends on the quality of the source data provided. |

*Prepared by SpectreTech Limited — Reliable Tech Solutions*
*Project Doxa Car Hire Management System · SP-DOXA-2026-001 · 27 July 2026*